

Rishikanth Manimeli

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Education

University of Massachusetts Amherst 01/2025 – 12/2026
MS in Computer Science | GPA: 4.0/4.0

- **Coursework:** Systems for Deep Learning, Reinforcement Learning, Machine Learning, Distributed and Operating Systems, Database Design and Implementation, Computer and Network Security, Cryptography

Indian Institute of Technology Patna 08/2018 – 05/2022
Bachelor of Technology in Electrical Engineering

Experience

Samsung India, Bangalore 06/2022 – 01/2025
Senior Software Engineer

- Optimized MAC Uplink scheduling for Cell on Wheels deployments across Telus and Verizon modems, increasing cell capacity and LTE throughput by 25%.
- Profiled and optimized core C++ system modules in a Linux environment to resolve CPU performance bottlenecks, reducing cycle consumption and improving throughput by 15%.
- Built automated CI/CD pipelines and regression test frameworks in Python and PowerShell, reducing manual log analysis from 25 minutes to a 10-minute scripted run.
- Drove architectural improvements to the modem scheduler; contributed 100+ commits and 300+ code reviews over two years across the modem software stack.

Projects

Agentic RAG Research Extern | Industry Experience at Adobe 01/2026 – 05/2026

- Built a multi-document question-answering pipeline using LlamaIndex and LangGraph, indexing research papers and internal documentation with sub-second retrieval latency.
- Designed a multi-hop reasoning benchmark on the Enron email dataset to detect data contamination in LLM training sets and evaluate retrieval accuracy across models.
- Created an agentic RAG workflow using tool calling to retrieve missing context mid-inference, reducing hallucination rates on internal benchmarks.

Stock Market Application | React, Python, AWS

- Built a fault-tolerant microservices trading platform on AWS using REST APIs; implemented LRU caching with server-push invalidations and leader-based replication for automatic failover.
- Implemented Raft and Multi-Paxos consensus algorithms to guarantee totally ordered writes across replicas and tolerate node failures using a majority quorum.
- Load-tested the system under concurrent multi-user traffic, validating consistent data integrity, low latency, and correct cache behavior at scale.

Database Implementation | Java

- Engineered a custom relational database management system (DBMS) from scratch in Java, implementing core storage components including a buffer pool manager, on-disk page management, and a heap file engine.
- Implemented a B+ tree index structure and an iterator-based query executor supporting Block Nested Loop (BNL) joins to optimize query execution.

Machine Learning Algorithms Implementation | Python

- Implemented core machine learning algorithms from scratch in Python – deep neural networks, decision trees, k-NN, and Naive Bayes – without external ML libraries.
- Built an end-to-end evaluation pipeline with hyperparameter tuning, cross-validation, and performance benchmarking across multiple datasets.

Technologies

Programming Languages: C++, Python, Java, Go, Rust, C, SQL, TypeScript, HTML/CSS, R

Software Development & Tools: AWS, Docker, Git, Linux, CI/CD, REST APIs

Frameworks & Libraries: React, Spring Boot, Qt, LangChain, LlamaIndex, LangGraph

Domains & Methodologies: Large Language Models, Agentic AI, Distributed Systems, Machine Learning, NLP

Leadership & Achievements

Course Grader & Project Mentor (COMPSCI 426): Led an 8-student team through Agile sprints on a scalable web systems project, evaluating code quality and architecture (Spring 2026).

Samsung Excellence Award: Recognized for exceptional contributions to system capacity improvement and core code optimization (April 2023).

Software Excellency Certification: Earned Samsung's Professional-level certification in algorithmic problem solving (October 2023).